Software Requirements Specification

FlatFinder: Housing Selection Assistant

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1. Project Mission Statement

The FlatFinder team will develop an interactive map-based web application that helps prospective HDB resale flat buyers in Singapore identify flats matching their budget, flat-type, and proximity preferences for MRT stations, bus stops, and schools. This project will be considered complete when the application has been tested and demonstrated to the lab supervisor, matching user-specified budget, flat-type, and proximity criteria against the confirmed data.gov.sg and OneMap datasets. This project supports prospective Singapore homebuyers in making faster, more informed resale flat decisions, replacing the current fragmented process of manually cross-referencing HDB resale listings, map applications, and transport directories with a centralised platform.

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2. Functional Requirements

1. The users shall be able to query the system to search for resale HDB flats.
    1.1. Users must query the system by specifying their requirements
        1.1.1. The system must allow the user to specify the minimum and maximum budget range in Singapore Dollars (SGD).
        1.1.2. The system must allow users to select one or more desired flat types (1 Room, 2 Room, 3 Room, 4 Room, 5 Room, Executive, Multi-Generation).
        1.1.3. The system must allow the users to specify an acceptable walking distance to transport nodes via slider control.
            1.1.3.1. The system must allow the users to specify an acceptable walking distance from bus stops.
            1.1.3.2. The system must allow the users to specify an acceptable walking distance from MRT Stations.

1.1.4. The system must allow users to assign an importance level from 1 (Not Important) to 5 (Essential) for each amenity category.

1.1.5. The system must allow the user to specify the minimum remaining lease

1.1.6. If a criterion is left blank, the system shall treat it as unconstrained

1.1.7. The system shall allow search via postal code/amenity name, geocoded through OneMap as a proximity anchor

1.2. The system must be able to calculate walking distances to surrounding amenities

1.2.1. the system shall query the OneMap Routing API to calculate the walking distances

1.2.1.1. If the OneMap API fails, the system shall display a user-facing error message

1.2.1.2. Upon API failure, the system shall preserve the last successfully rendered map state.

2. The system shall display an interactive map of Singapore.

2.1. The system shall allow users to pan and zoom the interactive map.

2.1.1. The interactive map shall display the total number of resale flats by area when zoomed out.

2.1.2. The interactive map shall display recommended flats when zoomed in

2.2. The interactive map must display the recommended resale flats matching the user's requirements

2.2.1. The recommended flats must be displayed on the map in the form of location pins

2.2.2. The system must calculate a fit score for each recommended flat

2.2.2.1. The system shall weight amenity categories with higher importance levels more heavily in the match percentage calculation.

2.2.3. The system shall display all recommended flats on a side panel

2.2.3.1. The system must report the details of each flat in the same format

2.2.3.1.1. The flat's displayed data must contain the street

2.2.3.1.2. The flat's displayed data must contain the block number

2.2.3.1.3. The flat's displayed data must contain the flat type

2.2.3.1.4. The flat's displayed data must contain the flat area

2.2.3.1.5. The flat's displayed data must contain the flat storey range

2.2.3.1.6. The flat's displayed data must contain the price in SGD

2.2.3.1.7. The flat's displayed data must contain the lease commence date

2.2.3.1.8. The flat's displayed data must contain the remaining lease
2.2.3.1.9. The flat's displayed data must contain the fit score.
2.2.3.2. The system will display recommended flats in descending order based on Fit Score.
2.2.3.3. The system must allow the user to reorder results by price, remaining lease, floor area or Fit Score
2.2.4. The system shall allow users to manage bookmarked resale flats
2.2.4.1. The system shall allow users bookmark resale flats
2.2.4.2. The system shall allow users to view their list of bookmarked resale flats.
2.3. The interactive map must display the nearest amenities within a 1km radius to selected resale flats
2.3.1. The interactive map must display the nearest bus stops using location pins
2.3.2. The interactive map must display the nearest MRT stations using location pins
2.3.3. The interactive map must display the nearest pre-schools using location pins
2.3.4. The interactive map must display the nearest hawker centres using location pins
2.3.5. The interactive map must display the nearest CHAS clinics using location pins
2.3.6. The system shall allow users to toggle visibility of each amenity category layer

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3. Non-Functional Requirements

ID	Category	Requirement
NFR 1	Performance	The system shall return search results within 5 seconds for a query against the resale HDB Flats dataset
NFR 2.1	Reliability & Availability	If the OneMap API or a data.gov.sg endpoint is unreachable, the system shall display an error message to the user.
NFR 2.2	Reliability & Availability	If the system is unable to find a flat based on the users' requirements, the system must display a message to prompt the user to adjust their requirements.

NFR 3.1	Usability	80% of first-time users must be able to complete the preference input flow within 2 minutes without external guidance.
NFR 3.2	Usability	All search criteria input controls must be visible in the sidebar without requiring the user to scroll, at a screen resolution of 1366×768 or larger.
NFR 4	Security	The system shall not require the user to submit any personally identifiable information (e.g. NRIC, full name, contact number) in order to perform a flat search.

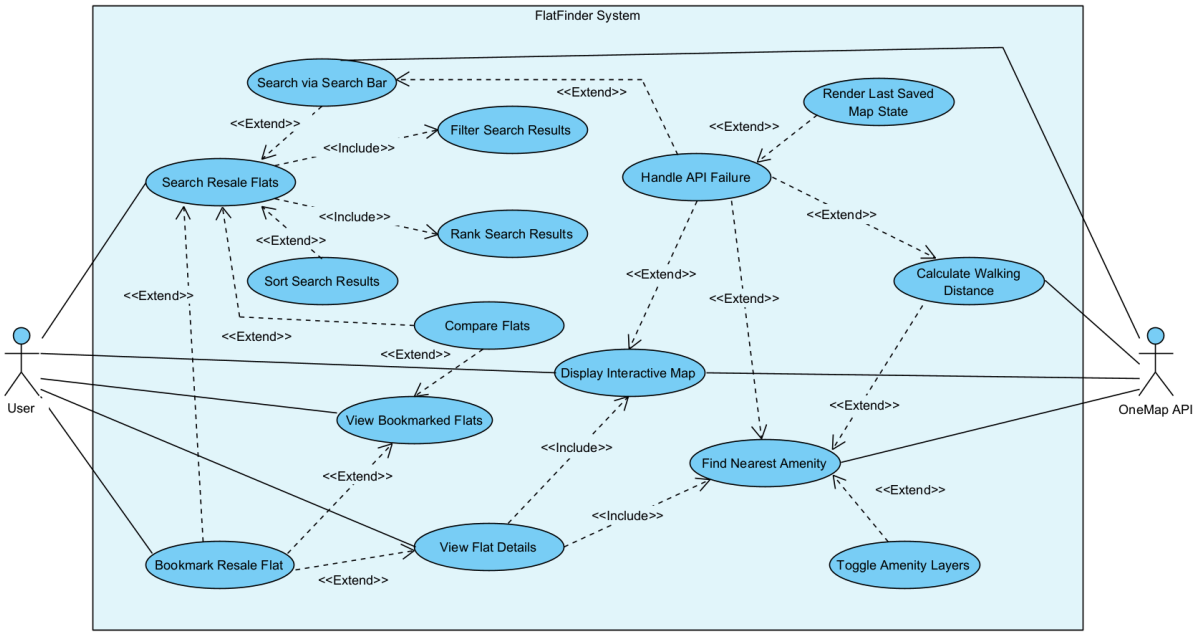
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4. Data Dictionary

Term	Description	Attributes	Relationships
User (Homebuyer)	A person using the application to find their ideal HDB resale flat.	Budget limits, desired flat sizes, acceptable walking distances, amenity importance scores (1–10 scale), a bookmarked flats list, and a selected sort order.	Creates Search Queries, saves and views Bookmarked Flats, selects a Sort Order for displayed results, and provides the baseline preferences for calculating Fit Scores.
Resale Flat	An individual HDB apartment listing pulled from the government dataset.	Address (street and block), flat type and size, floor level, price, lease details, and exact map coordinates.	Filtered by the User's search query, linked to nearby Transport Nodes and Amenities, assigned a unique Fit Score, and may be bookmarked by a User for future reference.
Transport Node	Public transit points, specifically MRT stations and bus stops.	Transit type (MRT or Bus), station/stop name, map coordinates, and calculated walking distance from the flat.	Linked to specific Resale Flats within 1km. Walking distance to a selected Transport Node is calculated via the OneMap Routing API (see Calculate Walking Distance). Evaluated against the User's acceptable walking distance preferences.
Amenity	Neighborhood facilities located within a 1km radius of a flat, such as	Amenity type, name, map coordinates, and straight-line distance from the flat.	Linked to specific Resale Flats to provide neighborhood context on the interactive map. Visibility per

	schools, hawker centres, and CHAS clinics.		amenity category can be toggled on the map without a new query (see Toggle Amenity Layers). Unlike a Transport Node, an Amenity's displayed distance is a straight-line estimate, not a OneMap-calculated walking distance.
Search Query	The active set of filters and preferences the user has dialed into the side panel.	The specific minimums, maximums, and any criteria left blank (treated as unconstrained); amenity importance weights (1–10 scale); and, optionally, a proximity anchor location resolved from a postal code or amenity name entered via the search bar.	Created by the User to dictate which Resale Flats appear on the map and how they are ranked. A search-bar location is geocoded via the OneMap API before being added as a proximity anchor.
Fit Score	A personalized rating showing exactly how well a specific flat aligns with the user's current search query.	The total overall score, plus individual weightings for price match, lease match, and proximity match per amenity category — each category weighted according to the User's 1–10 importance rating.	Calculated dynamically for each Resale Flat whenever the User adjusts their Search Query, including when a new proximity anchor is set via the search bar.
Bookmark	A saved preference that allows a user to view a flat they are interested in for future reference.	The time it was saved, the specific user's identifier, and the specific flat's identifier.	Acts as a bridge linking a User directly to a specific Resale Flat. A User's full set of Bookmarks is retrieved as a list via View Bookmarked Flats.
Sort Order	The field and direction the User has selected for reordering the displayed search results, independent of the underlying Search Query.	Selected field (price, remaining lease, floor area, or Fit Score) and direction (ascending or descending).	Applied by the User to the Resale Flats already returned by the active Search Query; does not change which flats are returned or their Fit Scores, only their displayed order.

5. Use Case Diagram



UC-01: Search Resale Flats

Use Case ID:	UC-01		
Use Case Name:	Search Resale Flats		
Created By:	Ryan Lim	Last Updated By:	Ryan Lim
Date Created:	25/8/26	Date Last Updated:	24/10/26

Actor:	User
Description:	Allows the user to specify budget, flat-type, walking distances, and lease criteria to retrieve a ranked list of matching HDB resale listings without submitting personally identifiable information.
Preconditions:	The system is active, the database is populated, and no personally identifiable information is required from the User.
Postconditions:	A ranked list of matching HDB resale flat listings ordered by Fit Score.
Priority:	High
Frequency of Use:	High
Flow of Events:	(In table below)
Alternative Flows:	UC-01.AC.1: User submits a location via UC-07 Search via Search Bar, in addition to or instead of the sidebar filters. UC-01.AC.2: User applies UC-10 Sort Search Results to reorder the displayed list by a field other than Fit Score.
Exceptions:	UC-01.EX.1: If zero records satisfy the query, the system displays a message prompting the User to adjust their requirements (NFR 2).
Includes:	UC-08: Filter Search Results, UC-09: Rank Search Results
Extends:	-
Extended By:	UC-05: Bookmark Resale Flat, UC-07: Search via Search Bar, UC-10: Sort Search Results
Special Requirements:	Search results must be returned within 5 seconds (NFR 1).
Assumptions:	All filter controls remain visible without scrolling at a screen resolution of 1366×768 or larger (NFR 3.2).
Notes and Issues:	-

Flow of Events

Step	Actor Action	System Response
1	User adjusts filter criteria (budget range, flat size type, minimum remaining lease) on the sidebar.	
2	User assigns priority weights (1–10) for proximity to Bus Stops, MRT Stations, Child Care Centres, Hawker Centres, and CHAS Clinics, then clicks "Find Matching Flats."	
3		System executes UC-08: Filter Search Results to query the resale flat dataset against the active filter criteria.

4		System executes UC-09: Rank Search Results to calculate a Fit Score and order the matching flats.
5		System displays the recommended flats in the results panel, in descending order of Fit Score.

UC-02: View Flat Details

Use Case ID:	UC-02		
Use Case Name:	View Flat Details		
Created By:	Ryan Lim	Last Updated By:	Ryan Lim
Date Created:	25/8/26	Date Last Updated:	22/9/26

Actor:	User
Description:	Displays detailed attributes for a selected resale flat and triggers rendering of its location and nearby amenities on the interactive map.
Preconditions:	UC-01 has executed and search results are displayed, OR the User is viewing flat location pins directly on the interactive map (UC-03) without having performed a search.
Postconditions:	1. A detailed flat metadata panel is rendered on the side panel. 2. The interactive map is centred on the selected flat and its nearby amenities are plotted.
Priority:	High
Frequency of Use:	High
Flow of Events:	(In table below)
Includes:	UC-03: Display Interactive Map, UC-11: Find Nearest Amenity
Extended By:	UC-05: Bookmark Resale Flat
Special Requirements:	Displayed flat data fields must match FR 2.2.3.1.1–2.2.3.1.9 (street, block, flat type, area, storey range, price, lease commence date, remaining lease, Fit Score).
Assumptions:	-
Notes and Issues:	Two entry points are supported: a listing selected from UC-01's search results, and a pin selected directly on the map (when the User navigates the map without first searching)

Flow of Events

Step	Actor Action	System Response
1	User selects a flat listing from the search results panel, OR clicks a flat location pin directly on the interactive map.	
2		System fetches the flat's metadata (street, block, price, remaining lease, Fit Score, etc.).
3		System executes UC-03: Display Interactive Map to render the flat's location (if not already active).
4		System executes UC-11: Find Nearest Amenity to plot nearby amenities within 1km of the flat.
5		System displays the fetched metadata in a standardised format on the side panel.

UC-03: Display Interactive Map

Use Case ID:	UC-03		
Use Case Name:	Display Interactive Map		
Created By:	Ryan Lim	Last Updated By:	Ryan Lim
Date Created:	25/8/26	Date Last Updated:	22/9/26

Actor:	User, OneMap API
Description:	Retrieves base map tiles and renders the interactive map of Singapore, centering on a specific flat's location when one is active.
Preconditions:	The system is active and the application interface is loaded.
Postconditions:	An interactive map view is displayed, supporting pan and zoom, and centred on a specific flat when applicable.
Priority:	High
Frequency of Use:	High
Flow of Events:	(In table below)
Alternative Flows:	UC-03.AC.1: If the User has bookmarked flats, bookmarked flats are also rendered as pins (see Step 5).
Exceptions:	UC-03.EX.1: If the OneMap API times out or is unreachable, the system triggers UC-06: Handle API Failure.
Includes:	-
Extended By:	UC-06: Handle API Failure
Special Requirements:	-
Assumptions:	-
Notes and Issues:	-

Flow of Events

Step	Actor Action	System Response
1	User navigates to the map view, OR the system passes specific coordinates from UC-02.	
2		System sends spatial coordinates to the OneMap API.
3	OneMap API returns the requested map tile assets.	
4		System renders the interactive map, allowing pan and zoom, centred on the flat if one is active.
5		System displays the total number of resale flats per area when zoomed out (FR 2.1.1), individual flat pins when zoomed in (FR 2.1.2), and if the User has any bookmarked flats.

UC-04: Calculate Walking Distance

Use Case ID:	UC-04		
Use Case Name:	Calculate Walking Distance		
Created By:	Ryan Lim	Last Updated By:	Ryan Lim
Date Created:	25/8/26	Date Last Updated:	24/10/26

Actor:	User, OneMap API
Description:	Calculates the walking distance from a selected resale flat to a specific nearby transport node or amenity chosen by the User, and draws the walking route on the interactive map.
Preconditions:	UC-11: Find Nearest Amenity is active and rendering amenity pins for the selected flat.
Postconditions:	A walking distance value and walking route are calculated and displayed for the selected amenity or transport node.
Priority:	Medium
Frequency of Use:	Medium
Flow of Events:	(In table below)
Alternative Flows:	-
Exceptions:	UC-04.EX.1: If the OneMap Routing API fails, the system triggers UC-06: Handle API Failure.
Includes:	-
Extends:	UC-11: Find Nearest Amenity
Extended By:	UC-06: Handle API Failure
Special Requirements:	-
Assumptions:	-
Notes and Issues:	The diagram omits a direct User actor line for this use case: the triggering pin-selection (Step 1) always happens within UC-11's active context, so the User's involvement is inherited through that relationship rather than drawn separately.

Flow of Events

Step	Actor Action	System Response
1	User selects a specific amenity or transport node pin.	
2		System queries the OneMap Routing API using the flat's coordinates and the selected amenity's coordinates.
3	OneMap Routing API calculates and returns the walking distance.	
4		System displays the calculated walking distance and draws the walking route on the map alongside the selected amenity or transport node.

UC-05: Bookmark Resale Flat

Use Case ID:	UC-05		
Use Case Name:	Bookmark Resale Flat		
Created By:	Ryan Lim	Last Updated By:	Ryan Lim
Date Created:	25/8/26	Date Last Updated:	22/9/26

Actor:	User
Description:	Adds or removes a specific flat listing from the User's active bookmark list, from the search results, the flat details page, or the bookmarked flats list itself.
Preconditions:	Flat listings are actively rendered.
Postconditions:	The flat's bookmark status is toggled and the "Saved" list and counter are updated accordingly.
Priority:	Low
Frequency of Use:	Medium
Flow of Events:	(In table below)
Alternative Flows:	UC-05.AC.1: If the listing is already bookmarked, selecting the icon removes the flat from the bookmark list and resets the icon state.
Exceptions:	-
Includes:	-
Extends:	UC-01: Search Resale Flats, UC-02: View Flat Details, UC-14: View Bookmarked Flats
Special Requirements:	Satisfies FR 2.2.4 (allow users to bookmark resale flats).
Assumptions:	Bookmarks persist for the User's active session only, since the SRS defines no User accounts or login.
Notes and Issues:	-

Flow of Events

Step	Actor Action	System Response
1	User selects the bookmark icon on a flat listing (in UC-01's search results, UC-02's flat details panel, or UC-14's bookmarked flats list).	
2		System toggles the flat's bookmark status: adds it to the bookmark list if not already present, or removes it if already bookmarked.
3		System updates the bookmark icon state and the counter.

UC-06: Handle API Failure

Use Case ID:	UC-06		
Use Case Name:	Handle API Failure		
Created By:	Ryan Lim	Last Updated By:	Ryan Lim
Date Created:	25/8/26	Date Last Updated:	22/9/26

Actor:	None (system-triggered)
Description:	Intercepts failed OneMap API requests, logs the failure internally, and displays a user-facing error message without crashing the application.
Preconditions:	A OneMap API call fails or times out during UC-03, UC-04, UC-07, or UC-11.
Postconditions:	The error is logged internally and the UI degrades gracefully, retaining the rest of the application's functionality.
Priority:	Low
Frequency of Use:	Low
Flow of Events:	(In table below)
Alternative Flows:	-
Exceptions:	-
Includes:	-
Extends:	UC-03: Display Interactive Map, UC-04: Calculate Walking Distance, UC-07: Search via Search Bar, UC-11: Find Nearest Amenity
Extended By:	UC-13: Render Last Saved Map State
Special Requirements:	Must satisfy NFR 2: an error message is displayed if the OneMap API endpoint is unreachable.
Assumptions:	-
Notes and Issues:	-

Flow of Events

Step	Actor Action	System Response
1		The OneMap API fails to respond within the expected threshold, or returns an error status.
2		System logs the API endpoint, timestamp, and error code to the internal system log.
3		System displays a dismissible error message to the User (e.g., "OneMap service currently unavailable") while retaining the rest of the application's functionality.
4		If the failure occurred while displaying the interactive map, the system executes UC-13: Render Last Saved Map State.

UC-07: Search via Search Bar

Use Case ID:	UC-07		
Use Case Name:	Search via Search Bar		
Created By:	Ryan Lim	Last Updated By:	Ryan Lim
Date Created:	22/9/26	Date Last Updated:	22/9/26

Actor:	User, OneMap API
Description:	Allows the User to search using a postal code or amenity name, geocoded via OneMap, as an alternative or supplement to the sidebar filters.
Preconditions:	UC-01: Search Resale Flats is active.
Postconditions:	The resolved location is set as a proximity anchor and the search results are updated accordingly.
Priority:	Medium
Frequency of Use:	Medium
Flow of Events:	(In table below)
Alternative Flows:	-
Exceptions:	UC-07.EX.1: If the OneMap API fails, the system triggers UC-06: Handle API Failure. UC-07.EX.2: If the query matches no location, the system displays a "no matching location found" message.
Includes:	-
Extends:	UC-01: Search Resale Flats
Extended By:	UC-06: Handle API Failure
Special Requirements:	-
Assumptions:	-
Notes and Issues:	-

Flow of Events

Step	Actor Action	System Response
1	User enters a postal code or amenity name into the search bar.	
2		System sends the query to the OneMap API for geocoding.
3	OneMap API returns the matching location(s) with coordinates.	
4		System sets the resolved location as a proximity anchor and re-executes UC-09: Rank Search Results.

UC-08: Filter Search Results

Use Case ID:	UC-08		
Use Case Name:	Filter Search Results		
Created By:	Ryan Lim	Last Updated By:	Ryan Lim
Date Created:	22/9/26	Date Last Updated:	24/10/26

Actor:	User
Description:	Lets the User constrain search results by budget range, flat size type, and minimum remaining lease.
Preconditions:	UC-01: Search Resale Flats is active.
Postconditions:	The active filter criteria are set for the current search query.
Priority:	High
Frequency of Use:	High
Flow of Events:	(In table below)
Alternative Flows:	-
Exceptions:	-
Includes:	-
Extends:	-
Extended By:	-
Special Requirements:	-
Assumptions:	-
Notes and Issues:	-

Flow of Events

Step	Actor Action	System Response
1	User adjusts the budget range (FR 1.1.1).	
2	User selects one or more desired flat types (FR 1.1.2).	
3	User specifies the minimum remaining lease (FR 1.1.5).	
4		Any criterion left blank is treated as unconstrained (FR 1.1.6).

UC-09: Rank Search Results

Use Case ID:	UC-09		
Use Case Name:	Rank Search Results		
Created By:	Ryan Lim	Last Updated By:	Ryan Lim
Date Created:	22/9/26	Date Last Updated:	24/10/26

Actor:	User
Description:	Calculates a Fit Score for each matching flat, based on the User's stated proximity importance weights, and orders the results accordingly.
Preconditions:	UC-08: Filter Search Results has executed and produced at least one matching flat.
Postconditions:	All matching flats are assigned a Fit Score and returned in descending order.
Priority:	High
Frequency of Use:	High
Flow of Events:	(In table below)
Alternative Flows:	-
Exceptions:	-
Includes:	-
Extends:	-
Extended By:	-
Special Requirements:	Must complete within the NFR 1 performance threshold (5 seconds), combined with UC-08's filtering step.
Assumptions:	-
Notes and Issues:	Distances used in the Fit Score calculation are straight-line distances stored in the database, not OneMap-calculated walking distances.

Flow of Events

Step	Actor Action	System Response
1	User assigns priority weights for proximity to Bus Stops, MRT, Hawker Centres, Child Care Centres, and CHAS Clinics via the sidebar sliders (FR 1.1.3, 1.1.4).	
2		System calculates a Fit Score for each matching flat, weighting amenity categories with higher importance levels more heavily (FR 2.2.2, 2.2.2.1).
3		System orders the matching flats in descending order of Fit Score (FR 2.2.3.2).

UC-10: Sort Search Results

Use Case ID:	UC-10		
Use Case Name:	Sort Search Results		
Created By:	Ryan Lim	Last Updated By:	Ryan Lim
Date Created:	22/9/26	Date Last Updated:	22/9/26

Actor:	User
Description:	Lets the User reorder the displayed search results by a field other than Fit Score.
Preconditions:	UC-01: Search Resale Flats has returned at least one result.
Postconditions:	The displayed results are reordered by the selected field.
Priority:	Low
Frequency of Use:	Medium
Flow of Events:	(In table below)
Alternative Flows:	-
Exceptions:	-
Includes:	-
Extends:	UC-01: Search Resale Flats
Extended By:	-
Special Requirements:	-
Assumptions:	-
Notes and Issues:	-

Flow of Events

Step	Actor Action	System Response
1	User selects a sort field from the "Sort by" dropdown (Price, Remaining Lease, Floor Area, or Fit Score).	
2		System reorders the displayed results by the selected field.

UC-11: Find Nearest Amenity

Use Case ID:	UC-11		
Use Case Name:	Find Nearest Amenity		
Created By:	Ryan Lim	Last Updated By:	Ryan Lim
Date Created:	22/9/26	Date Last Updated:	22/9/26

Actor:	User, OneMap API
Description:	Discovers and plots amenities and transport nodes within a 1km radius of a selected resale flat.
Preconditions:	UC-02: View Flat Details is active for a specific flat.
Postconditions:	Nearby amenities within 1km of the flat are plotted as pins on the interactive map.
Priority:	High
Frequency of Use:	High
Flow of Events:	(In table below)
Alternative Flows:	-
Exceptions:	UC-11.EX.1: If the OneMap API fails, the system triggers UC-06: Handle API Failure.
Includes:	-
Extends:	-
Extended By:	UC-04: Calculate Walking Distance, UC-06: Handle API Failure, UC-12: Toggle Amenity Layers
Special Requirements:	Amenity search radius is fixed at 1km (FR 2.3).
Assumptions:	-
Notes and Issues:	-

Flow of Events

Step	Actor Action	System Response
1		System queries the OneMap API for amenities within 1km of the flat's coordinates.
2	OneMap API returns matching amenity locations.	
3		System drops location pins for the nearest bus stops, MRT stations, pre-schools, hawker centres, and CHAS clinics (FR 2.3.1–2.3.5).
4	User selects a specific amenity or transport node pin.	
5		System executes UC-04: Calculate Walking Distance for the selected pin.

UC-12: Toggle Amenity Layers

Use Case ID:	UC-12		
Use Case Name:	Toggle Amenity Layers		
Created By:	Ryan Lim	Last Updated By:	Ryan Lim
Date Created:	22/9/26	Date Last Updated:	22/9/26

Actor:	User
Description:	Lets the User show or hide specific amenity categories on the interactive map without re-querying the underlying dataset.
Preconditions:	UC-11: Find Nearest Amenity is active and amenity pins are rendered.
Postconditions:	Only the selected amenity categories are visible on the map.
Priority:	Low
Frequency of Use:	Medium
Flow of Events:	(In table below)
Alternative Flows:	-
Exceptions:	-
Includes:	-
Extends:	UC-11: Find Nearest Amenity
Extended By:	-
Special Requirements:	-
Assumptions:	-
Notes and Issues:	Toggling a layer only shows or hides pins already fetched by UC-11; it does not call the OneMap API again.

Flow of Events

Step	Actor Action	System Response
1	User selects or deselects an amenity category toggle (e.g., Bus Stops, MRT Stations, Schools, Hawker Centres) in the map's layer control.	
2		System shows or hides the pins for that category within the currently rendered amenity set.

UC-13: Render Last Saved Map State

Use Case ID:	UC-13		
Use Case Name:	Render Last Saved Map State		
Created By:	Ryan Lim	Last Updated By:	Ryan Lim
Date Created:	22/9/26	Date Last Updated:	22/9/26

Actor:	None (system-triggered)
Description:	Re-displays the last successfully rendered map state after an API failure occurs while displaying the interactive map.
Preconditions:	UC-06: Handle API Failure has been triggered while UC-03: Display Interactive Map was active.
Postconditions:	The map reverts to and displays its last successfully rendered state.
Priority:	Low
Frequency of Use:	Low
Flow of Events:	(In table below)
Alternative Flows:	-
Exceptions:	UC-13.EX.1: If no prior map state exists (e.g., the first load fails), the system displays a blank map together with the error message from UC-06.
Includes:	-
Extends:	UC-06: Handle API Failure
Extended By:	-
Special Requirements:	Satisfies FR 1.2.1.2 (preserve the last successfully rendered map state on API failure).
Assumptions:	-
Notes and Issues:	The last rendered map state (tiles, pins, zoom/pan position) is cached client-side.

Flow of Events

Step	Actor Action	System Response
1		System retrieves the last successfully rendered map state (tiles, pins, zoom/pan position) from local cache.
2		System renders the cached map state in place of the failed request.

UC-14: View Bookmarked Flats

Use Case ID:	UC-14		
Use Case Name:	View Bookmarked Flats		
Created By:	Ryan Lim	Last Updated By:	Ryan Lim
Date Created:	22/9/26	Date Last Updated:	22/9/26

Actor:	User
Description:	Displays the list of flats the User has bookmarked during the active session.
Preconditions:	The User has bookmarked at least one flat (UC-05).
Postconditions:	The bookmarked flats list is displayed.
Priority:	Medium
Frequency of Use:	Medium
Flow of Events:	(In table below)
Alternative Flows:	UC-14.AC.1: If no flats are bookmarked, the system displays an empty-state message instead of a list.
Exceptions:	-
Includes:	-
Extends:	-
Extended By:	UC-05: Bookmark Resale Flat
Special Requirements:	-
Assumptions:	-
Notes and Issues:	-

Flow of Events

Step	Actor Action	System Response
1	User selects the "Saved" button/counter.	
2		System retrieves the User's bookmarked flats from session memory.
3		System displays the bookmarked flats in a list, OR pins on the interactive map.

UC-15: Compare Flats

Use Case ID:	UC-015		
Use Case Name:	Compare Flats		
Created By:	Ryan Lim	Last Updated By:	Ryan Lim
Date Created:	24/10/26	Date Last Updated:	24/10/26

Actor:	User
Description:	Allows the user to collect two or three flats into a session tray and compare them side by side using an adjustable weight model.
Preconditions:	The user has search results (UC-01) and/or saved flats (UC-14), and at least two flats have been added to the compare tray.
Postconditions:	A comparison page shows each flat's attributes and a factor-by-factor score breakdown, the highest-scoring flat is marked as the winner.
Priority:	Medium
Frequency of Use:	Medium
Flow of Events:	(In table below)
Alternative Flows:	UC-15.AC.1: Without an active search, the user compares two or three saved flats directly. Price, remaining lease and floor area use absolute reference scales, and every factor defaults to weight 5.
Exceptions:	-
Includes:	-
Extends:	UC-01: Search Resale Flats, UC-14: View Bookmarked Flats
Extended By:	-
Special Requirements:	The comparison total is 100 × Σ(weight × score) / Σ weight over the non-ignored factors. With an active search, price and remaining lease reuse the Fit Score's normalisation and fixed weights, so the default comparison total equals the Fit Score.
Assumptions:	The compare tray is stored in the session and lasts for the session only.
Notes and Issues:	Ties are broken by lower price, then longer remaining lease.

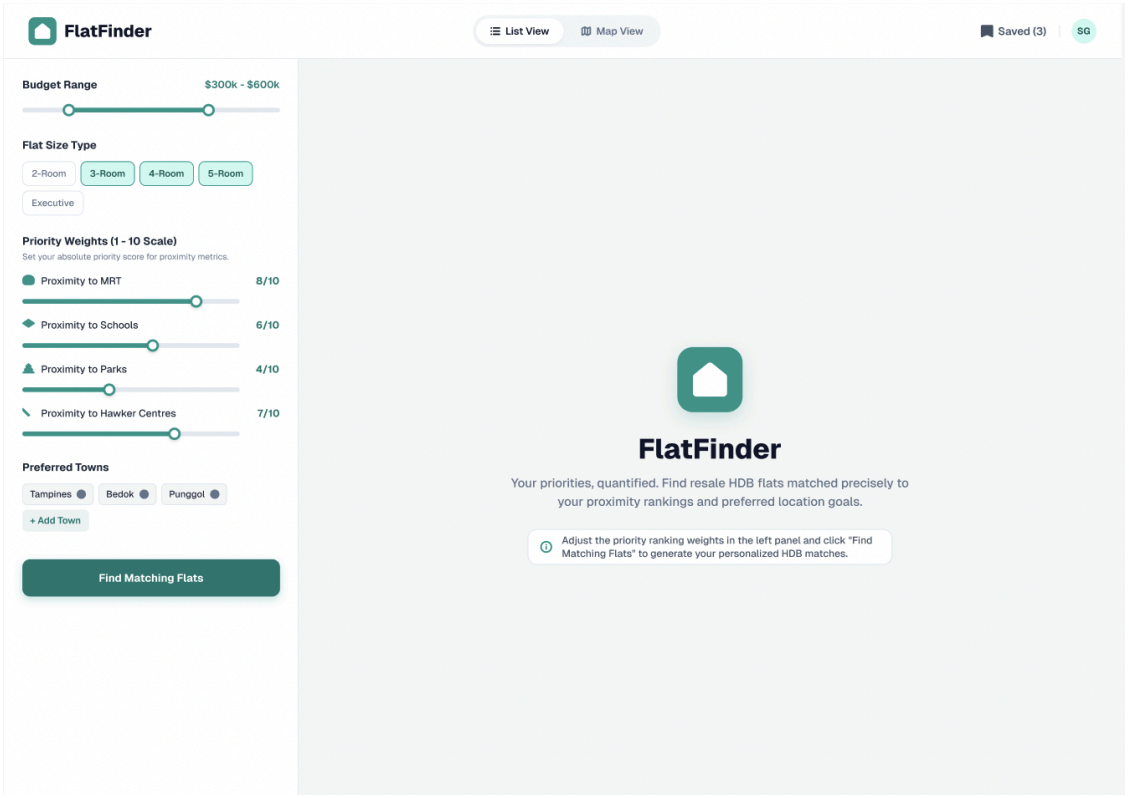
Flow of Events

Step	Actor Action	System Response
1	User adds a home to the compare tray from Results or Saved (a block contributes its best-ranked flat).	
2		System stores the flat's id in the session tray (capped at three) and updates the "Compare (n)" counter.
3	User opens the Compare page.	
4		System renders one column per home, with its attributes and a factor-by-factor score breakdown.

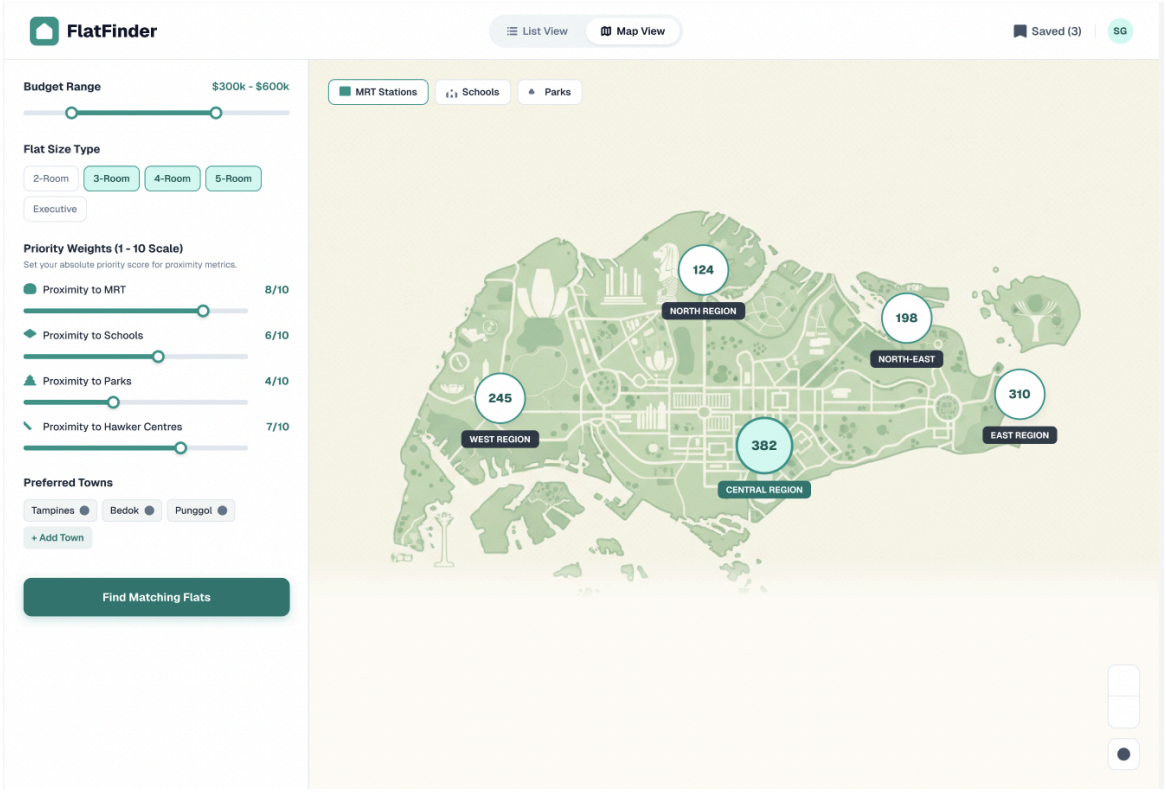
5	User adjusts factor weights (0 = ignore, 1–10 = importance).	
6		System recomputes each home's total and marks the winner

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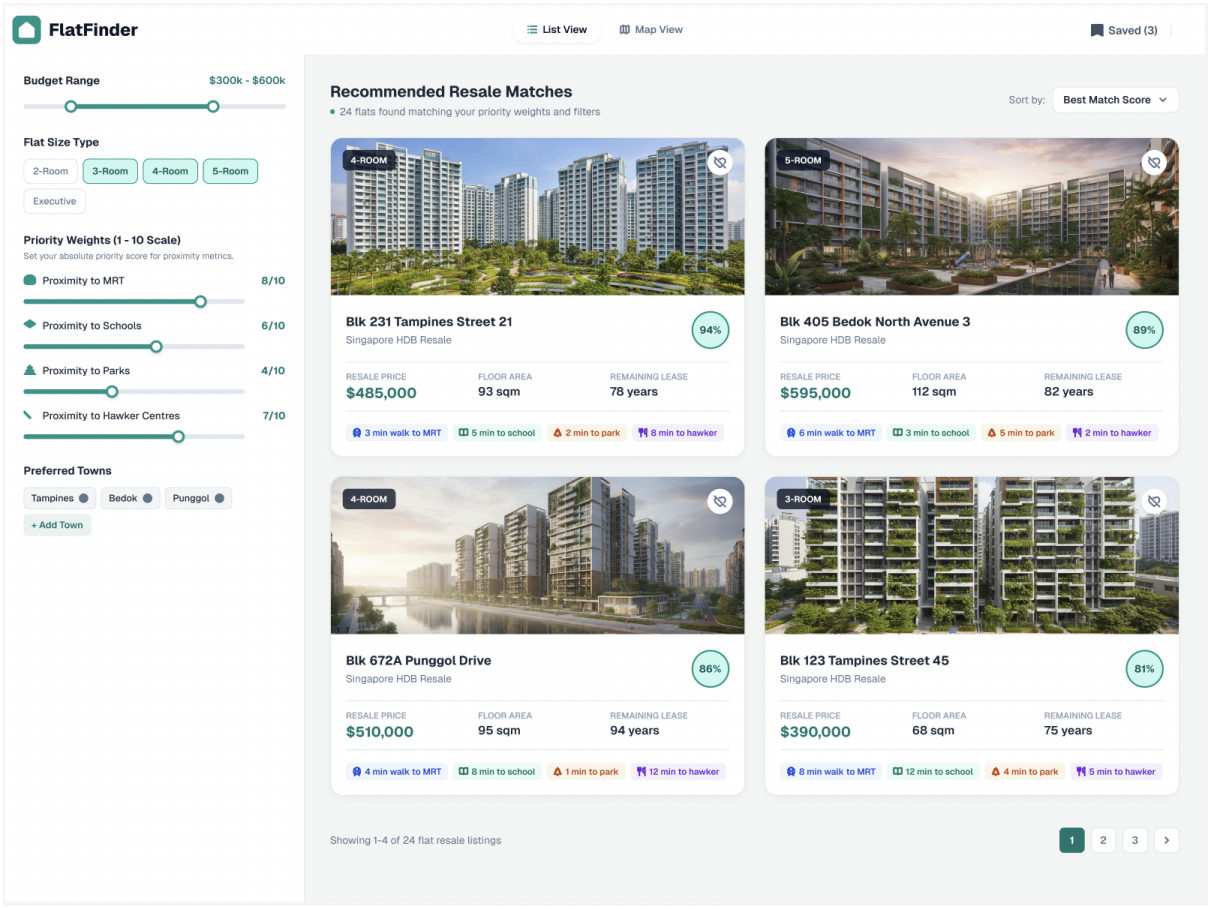
6. UI Mockups



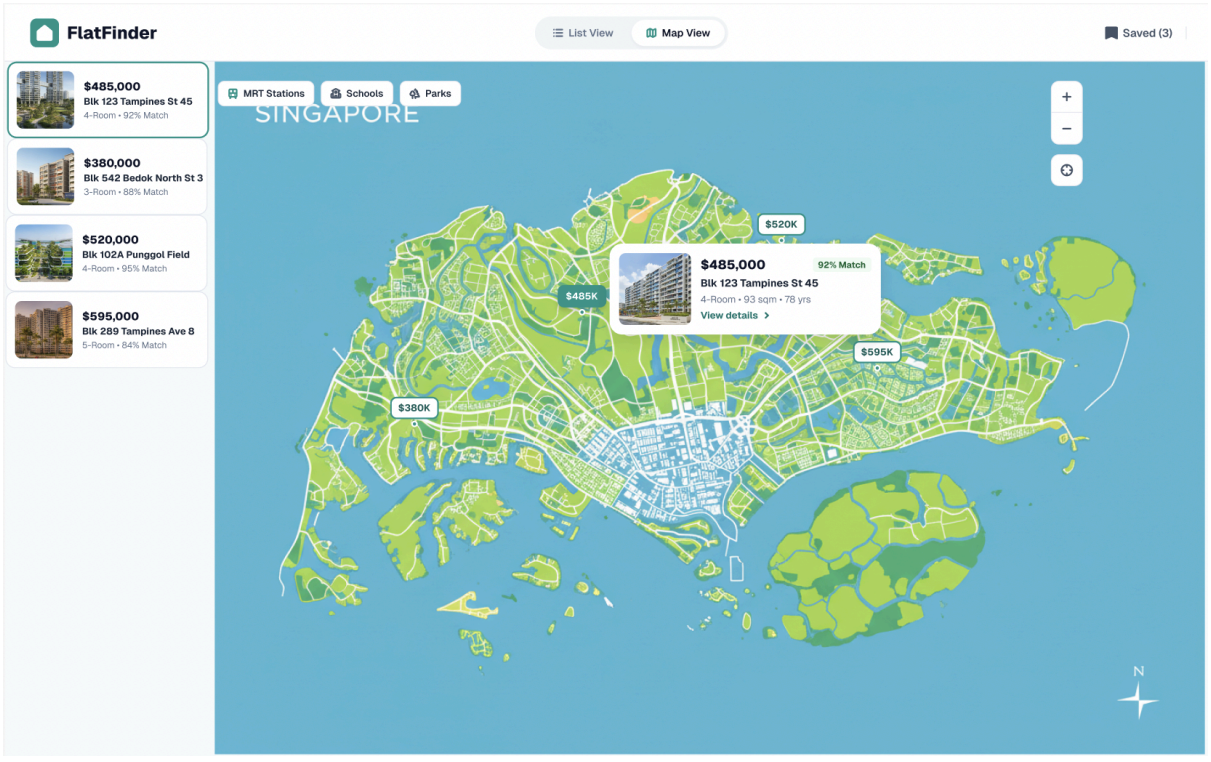
Start Page (List View)



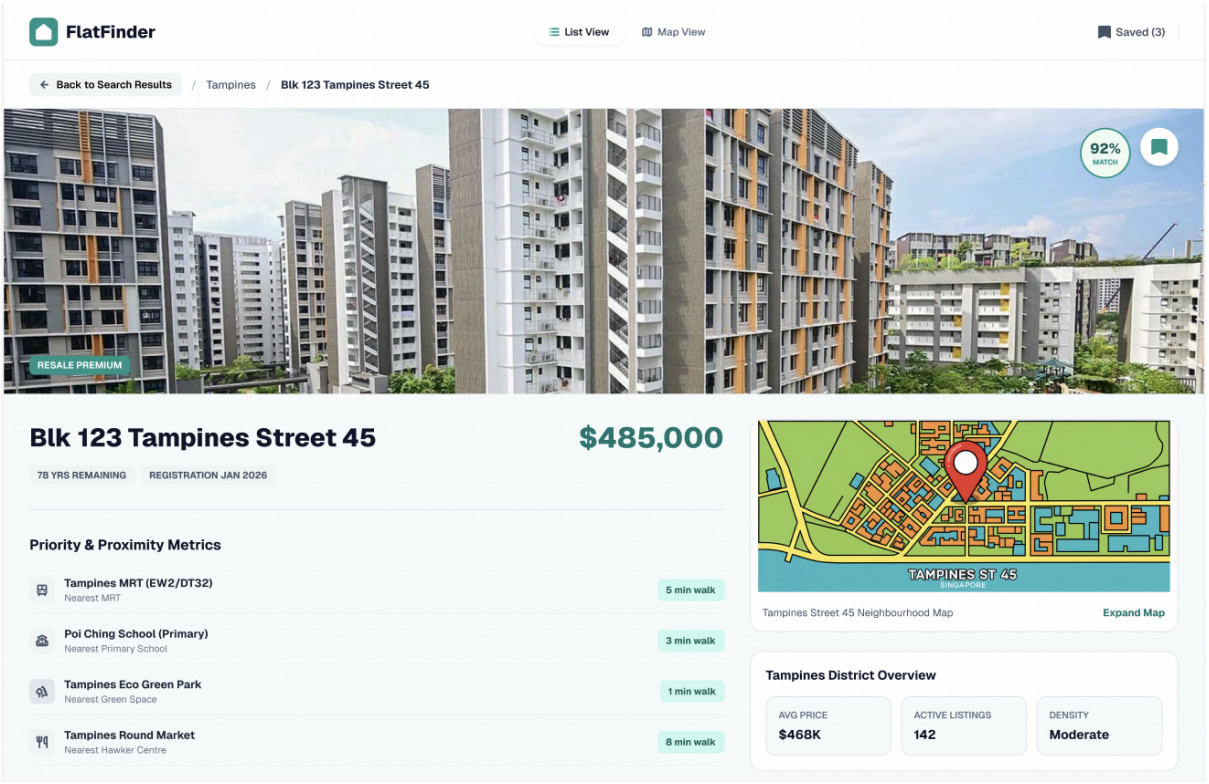
Start Page (Map View)



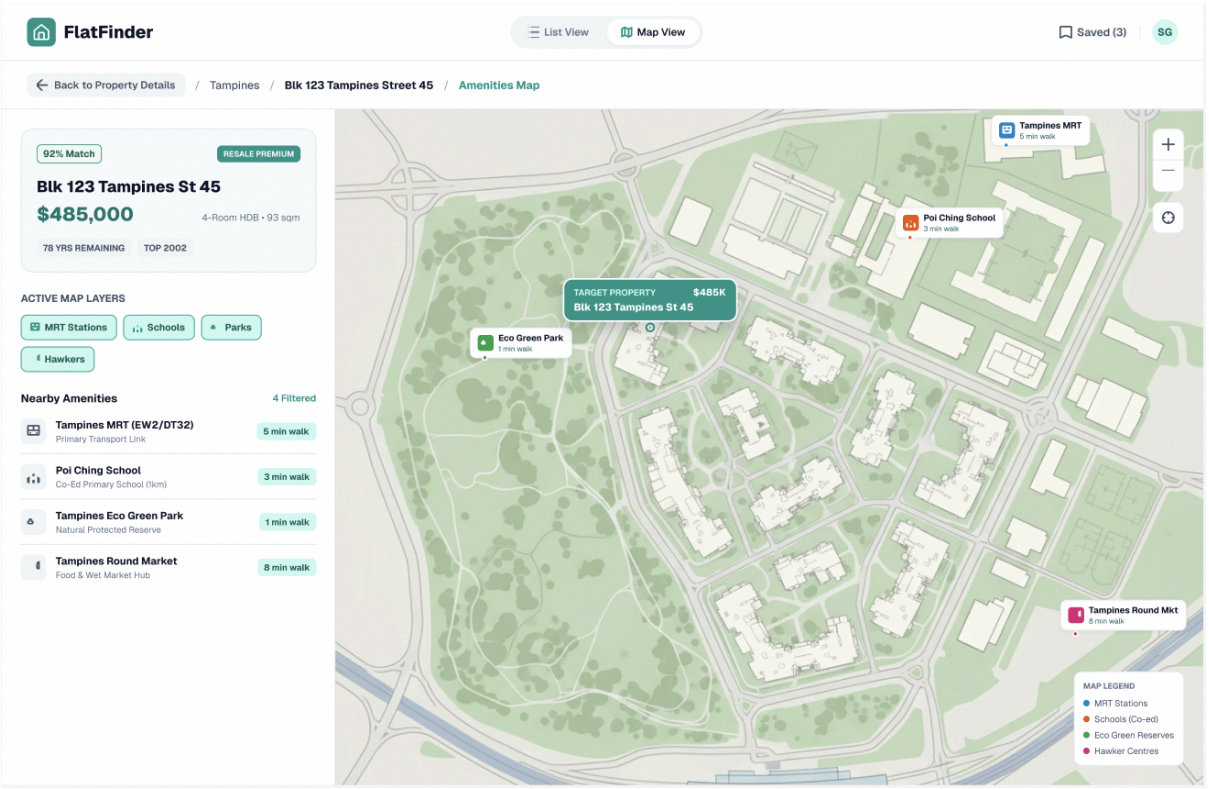
Results Page (List View)



Results Page (Map View)



Property Details Page (List View)



Property Details Page (Map View)

Explanation of UI Mockups:

- When a user first opens the website, they will see the start page. They can then adjust the filters on the left sidebar as per their preferences and click on “Find Matching Flats”. The app will then select flats that match their preferences and show it to them in the results page. The user can look through the flats and click on one of them to view the details of the listing. This will then bring them to the property details page, where the user will be able to see the details of the flat and nearby amenities. At any point in this process, the user is able to toggle between the list view and map view.
- Alternatively, the user can toggle to the map view at the start page. They can then zoom in to a particular region and look at flat listings in that area, and click on one of those listings to view the property details. This will bring them directly to the property details page (map view), bypassing the results page.

References

Dataset Links

Resale Flats 2017 Onwards

https://data.gov.sg/datasets/d_8b84c4ee58e3cfc0ece0d773c8ca6abc/view

Town / planning area boundaries (for the spatial join)
https://data.gov.sg/datasets/d_2cc750190544007400b2cfd5d7f53209/view
Used for UX enhancement for interactive map. Highlights areas and draws borders around it

MRT station locations
https://data.gov.sg/datasets/d_b39d3a0871985372d7e1637193335da5/view
*(LTA MRT Station Exit — used for **FR3.2**)*

Bus stop locations
https://data.gov.sg/datasets/d_3f172c6feb3f4f92a2f47d93eed2908a/view
*(LTA Bus Stop — used for **FR3.3**)*

Child Care Services Locations

https://data.gov.sg/datasets?query=&formats=GEOJSON&resultId=d_5d668e3f544335f8028f546827b773b4

Hawker Centre Locations

https://data.gov.sg/datasets?query=&formats=GEOJSON&resultId=d_4a086da0a5553be1d89383cd90d07ecd

CHAS Clinics

https://data.gov.sg/datasets/d_548c33ea2d99e29ec63a7cc9edcccedc/view
Excludes public polyclinic and hospitals